always be the case when crank setting is correct, whether the eccentric rod shows long or short.

If the eccentric rod is $\frac{1}{8}$ inch too short

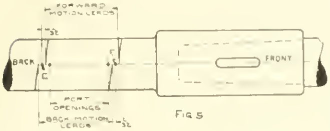

Fig 5

and the crank setting correct, the full gear lead lines will come as shown by Fig. 5:

After lengthening the rod $\frac{1}{8}$ inch you will have the condition shown by Fig. 4.

Questions and Answers for Possible Failures of the Baker Gear.

Q.—Is the Baker gear a direct or an indirect motion?

A.—It is direct, going ahead for an inside admission and indirect backing up, and just the opposite for the outside admission type.

Q.—What are the advantages of the Baker gear over other valve gears?

A.—It is an outside gear having no links or eccentrics. The bearings are all pins and bushings, which makes it very easy to repair. It is also standard, regardless of size or class of engine.

Q.—What means are provided to keep the gear properly lubricated?

A.—All bearings are provided with a pocket or cavity cast in casting, which does away with the use of oil cups.

Q.—What parts of the Baker gear take the place of the link which is used by the Stephenson or Walschaert motion?

A.—The radius bars and reverse yoke.

Q.—What relation to the main pin is the eccentric crank set?

A.—The eccentric crank always follows the main pin.

Q.—Should the eccentric rod or eccentric crank break, how is the engine put in condition to proceed?

A.—The disabled side can have lap and lead travel and a port opening equal to the lead for all cutoffs. First block the bell crank by using a U-bolt (which should be provided) in the holes placed in the gear frame for this purpose. Throwing reverse lever in mid-gear will help to get bell crank in position to block. Second, take down broken parts. Third knock out back pin of short reach rod and throw reverse yoke in forward motion against gear frame.

Q.—What is done should a gear connection rod break?

A.—Do the same as for a broken eccentric rod or crank.

Q.—What is done should the upper part of gear connection rod break?

A.—If break is close to the middle pin, do the same as for a broken eccentric rod and also tie lower end of gear connection rod to keep it from swinging. If break is near the top and below the jaw, first block the bell crank and wire the connection rod fast to radius bars. If break is through top jaw, do the same as for broken eccentric rod.

Q.—What is done should a radius bar break?

A.—Do the same as for broken eccentric rod.

Q.—If the horizontal arm of bell crank should break?

A.—Same as broken eccentric rod.

Q.—What is done should the vertical arm or bell crank break?

A.—Take down union link, combination lever and valve rod, then block valve over ports by using set screw in valve stem crosshead provided for that purpose.

Q.—Should you break crosshead arm or union link, what would you do?

A.—If rod be provided to secure lower end of combination lever to guide yoke, remove broken parts and proceed with full train, working engine at long cutoff. Otherwise would remove broken parts, combination lever and valve rod, cover ports and proceed on one side.

Q.—What do you do if a union link should break?

A.—Same as for a broken crosshead arm.

Q.—What is done if a combination lever should break?

A.—Tie combination lever plumb same as for a broken crosshead arm, if it is possible. If not possible, take down the combination lever and valve rod and cover the ports.

Q.—What is done if a valve rod breaks?

A.—Take down the broken parts and cover ports, leaving the rest of the gear intact.

Q.—What is done if a reverse yoke breaks?

A.—If lugs for holding reach rod break, block yoke securely at whatever cutoff you wish to work the engine and take down the short reach rod. If break is below the lugs, do the same as for broken eccentric rod.

Q.—What do you do if reach rod should break?

A.—If short reach rod breaks block the yoke at cutoff desired and wire fast so it cannot move. If main reach rod breaks, block between tumbling shaft arm and crosstie brace, wiring same securely.

Q.—What is done if the engine breaks down other than valve gear?

A.—In this case do the same as for any other valve gear.

To Keep Machinery from Rusting.

Dissolve one ounce of camphor in one pound of melted lard; take off the scum and mix in as much fine black lead as will give it a color. Clean the machinery and smear it with the mixture.

Meeting of the Executive Committee of the International Railway General Foremen's Association.

An important meeting of the officers and members of the Executive Committee of the International Railway General Foremen's Association will be held at the Sherman Hotel, Chicago, Tuesday, December 8, 1914, at 10 a. m. As matters of great importance to the organization are to be considered it is earnestly desired that all concerned will make an effort to be present.

The work of the association during the year is of the most interesting kind, and the various committees are already busy at work on the subjects assigned to them, and it is the determination of the officers and members of the Executive Committee that no effort will be spared to present in the best form all the facts in relation to the subjects under consideration that can be collected. With the excellent opportunities that the members have of making observations in actual practice and comparing data, valuable results may be safely expected.

Mr. William Hall, the secretary-treasurer of the association, announces that his address is now 1126 West Broadway, Winona, Minn.

Minimum Wage Law Declared Unconstitutional.

Holding the minimum wage law passed by the 1913 Minnesota Legislature to be unconstitutional, Judge Catlin, in the State District Court, handed down a decision ordering a temporary injunction against State Auditor Iverson and the members of the Minimum Wage Commission from expending further money. The ruling also suspends the order of the commission fixing a minimum wage for women and minor workers, which was to become effective today.

Judge Catlin's order makes the minimum wage law ineffective until the case has been tried on its merits. His decision will be appealed to the Supreme Court.

"The actual working of the law would be apt to increase immorality if morals are dependent upon wages," Judge Catlin asserted in the ruling.

The law was held to be unconstitutional because it delegated the legislative power to an appointive commission and placed in that commission a discretion as to whether there should be a minimum wage. "When the State [illegible] lawfully become a party [illegible] until the form of government has been entirely changed," Judge Catlin declared.

The second constitutional ground on which the court bases the decision was the abridgment of the right of the individual to contract. The law requires [illegible] employer [illegible] to the court.

A.—The boiler should be filled up while the fire is burning brightly at end of trip.

Q. 23.—Explain the use of the blow-off cock?

A.—The blow-off cock is to be used in clearing the water of foreign matter that causes it to foam, and should be used when standing still after filling boiler with water and the fire is burning brightly, blowing out about two or three gauges of water, depending on height of water above crown sheet. The injectors may be kept at work while the blow-off cock is open, if the fire is in condition to prevent cooling effect.

Q. 24.—How should the draught be distributed in the fire-box?

A.—Evenly all over the entire grate surface.

Q. 25.—Name the various adjustable appliances in the front end by which the burning of the fire is regulated?

A.—Nozzle tips, petticoat or draft pipe, and diaphragm plates.

Q. 26.—What does it indicate when the exhaust issues strongest from one side of the stack?

A.—It indicates that the draft pipe is out of plumb, caused by loosening of supports, or it may be that the nozzles are out of plumb, so that the exhaust does not strike the center of the smokestack.

Q. 27.—Why is it important that there be no holes through the smoke-box sheets or the front end and none in smoke-box seams or joints?

A.—It would spoil the vacuum formed by the forced draught and injure the steaming qualities of the engine.

Q. 28.—Describe the principles upon which the inspector works?

A.—A jet of high velocity steam passing through a small orifice strikes the water and is combined with the water, and condensed into water giving to the water part of its velocity, carrying the water along through smaller openings, and in that manner increasing its velocity, to such an extent that it has a force sufficient to overcome the boiler check and pass into the boiler.

Q. 29.—What is the difference between a lifting and a non-lifting injector?

A.—A lifting injector must have parts that will raise the water up into the body of the injector which is placed above the water level, and such other parts as will force the water into the boiler, while the non-lifting injector being placed below the water level has only such parts as will force the water into the boiler, the water flowing into it through force of gravity.

Q. 30.—Will injector work with a leak between injector and tank? Why? Will it prime?

A.—No. Because the air will spoil the vacuum. No. It will not prime for the same reason.

Q. 31.—If it primes good but breaks when the steam is turned on, where would you look for the trouble?

A.—Would look for some obstruction in the delivery tubes, or in discharge pipe, or boiler check might be stuck shut, or feed pipe might be stopped up or leaking.

Q. 32.—If it will not prime, where would you inspect to find the trouble?

A.—In the supply pipe, either the feed water too hot, a leak, or an obstruction in the feed pipe or strainer.

Q. 33.—Will injector prime if checks leak badly or are stuck up? If injector throttle leaks badly?

A.—No. Because the hot water and steam getting into the feed pipe spoils the vacuum formation.

Q. 34.—If steam or water shows at the overflow pipe when injector is not working, how do you tell whether it is boiler check or throttle leaking?

A.—With the boiler check leaking considerable water will show at the overflow pipe, and with throttle leaking steam will show at overflow pipe. To be sure you may test by closing the main steam valve at the fountain head and in that manner shut off the steam from the injector, or by forcing the boiler check to its seat with the combination boiler check and stop the flow of water from boiler. Another way is to open the pet cock in discharge pipe and if water comes from it with force it indicates that the check is stuck open.

Q. 35.—Will injector prime if primer valves leak? Will it prevent its working?

A.—Yes, it will prime. It will not prevent it working if primer valve leaks.

Q. 36.—Will an injector work if air cannot get into tank as fast as water is taken out?

A.—No. It will not work.

Q. 37.—Will an injector work if all of the steam is not condensed by the water?

A.—It will not work as it should, and if it works at all it will waste a great amount of water at the overflow. Generally it will not work at all.

Q. 38.—What precaution should be taken to prevent injectors from freezing when not being used?

A.—It should be changed into a heater to keep the feed pipe and discharge pipe from freezing.

Q. 39.—How would you set a heater on the various types of injectors in use on this road?

A.—On the twin Hancock injectors it is necessary to close the overflow valve and open the throttle valve enough to furnish sufficient steam to circulate in the delivery and feed pipes, and the drain cock in delivery pipe must be opened a little, and tank valve nearly closed. On other types of injectors after closing the overflow valves, close the main steam valve and open the injector throttle valve wide, then regulate the steam by opening the main steam valve the necessary amount to provide the circulation.

Q. 40.—Is any more water used when a boiler foams than when the water is solid?

A.—Yes. Because water is being carried out through cylinders with the steam.

Q. 41.—How should lubricator be filled?

A.—All feeds should be closed, water valve closed, condensation drained from the oil chamber, and then the oil chamber should be filled to within about one-half an inch of the top. (It should never be filled to overflowing because the oil expands with the heat and space should be allowed for the expansion.) The water valve should be opened as soon as the plug is put in the filling opening, to allow the condensation to raise the oil to top of oil well.

Wise Don'ts.

When a new device is added to a locomotive many men begin at once to expect performances from it that the inventor never thought of. The superheater has proved very efficient, but we greatly admire the list of "don'ts" covering its performance suggested by a committee of the Foundry Engineers' Association, of which Mr. J. W. Hardy was chairman. The list reads:

"Don't expect too much of the superheater; it is not intended to overcome blows or stop steam leaks or square valves, and it is like some children—won't keep itself clean.

"Don't forget when switching that there is more steam between the throttle and cylinders with the superheater than with the saturated steam engine—the superheater holds some.

"Don't carry water too high just because you don't hear any in the smoke stack. You might be using your superheater to boil water instead of heating steam.

"Don't think because your engine steams that you are getting the full value of the superheat; your engine may not be calling for the capacity of your boiler.

"Don't close your throttle entirely on road engines until you get to going quite slow; your cylinder lubrication will be much better.

"Don't fire your coal too wet; it won't clinker so badly if reasonably dry. The more you rake the fire the more the flues will stop up. There are only two reasons why a fire should be raked, one, because too much coal is used, and the other because it is not put in the right place.

"While there is a great difference in coal there is not as much difference as in what YOU are able to get out of it. They tell us of the high number of heat units or B. T. U.'s in certain coal; what does that amount to, to us, if we are not able to catch them harness them up and use them to our advantage?"

These don't apply with equal force to the handling of engines that are not equipped *with superheaters.*